ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 9



UNIT 6
ICT and Society

Student Workbook

"The King of ICT"

CHAMEEN RATHNASENA

Senior Lecturer in IT
Motivational Speaker
BSc (Sp) Computer Science - University of Peradeniya
PGDE - Open University of Sri Lanka
MIT (Reading) - University of Sabaragamuwa

STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain how ICT has changed our society, and what e-learning is.
- [ ] Name the ICT services of schools, hospitals, the public service and shops.
- [ ] Explain the digital divide.
- [ ] Name ten harmful electronic waste (e-waste) items.
- [ ] Say how the toxic poisons in e-waste harm people.
- [ ] Use the 3R technique: Reduce, Reuse and Recycle.
- [ ] Name ten computer related jobs and say what each person does.

THREE RULES FOR THE COMPUTER ROOM

- **Always** use the Internet with your teacher, and only safe websites.
- **Never** open, break or burn old electronic items. They hold toxic poisons.
- **Always** switch off the computers and the projector after the lesson.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the activities (a survey, a poster, a plan ...). Last you **write** the exercises in this book.

Colour the small pictures when you finish your work early.



Colour me!

THE WORDS YOU WILL LEARN

Society	Efficiency	e-learning	e-health
e-governance	e-commerce	Online shopping	Gazette
Digital divide	Developing	e-waste	Motherboard
Cathode Ray Tube	CFL bulb	Electric cell	Toxic
Mercury	Lead	Reduce	Reuse
Recycle	Programmer	Database	IT Consultant

"ICT changes the world every day. Learn it well, and use it to make your society better."

- THE KING OF ICT

CHAPTER 1

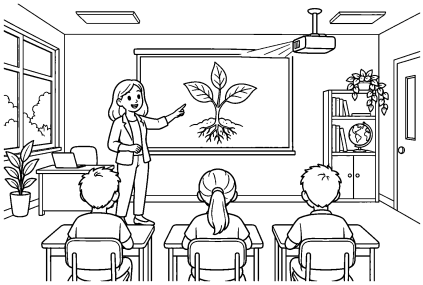
ICT Changes Our Society

? A NEW WAY TO LEARN

"Today we saw a Science video and photos for History on a computer. I love learning now!"

"Yes, that is e-learning: teaching using ICT and the Internet."

ICT has changed every field. Meet the services we get with ICT, and the digital divide.



6.1 ICT changes our society

Information and Communication Technology (**ICT**) has brought about great changes everywhere, in every field. The use of ICT has helped us to complete tasks more **efficiently** and **quickly**.

Even the methods of teaching have changed. A video in the Science lesson and photos for History on a computer help pupils to remember the lessons. This is **e-learning**: teaching using ICT and the Internet.

6.2 ICT services in four fields

These are some of the services we get with ICT (textbook page 76):

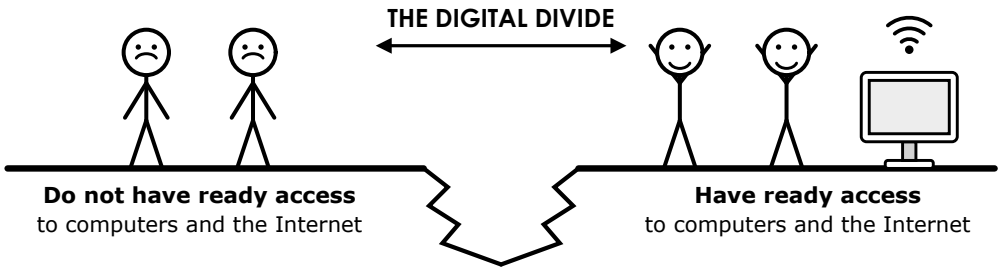
	Field	Services we get with ICT
	**School e-learning**	• Clearer explanation with pictures and video clips • Collecting information through the Internet • Dialogue and messaging with other schools • The facility to learn anytime from anywhere
	**Hospital e-health**	• Use of computer based modern equipment to diagnose illnesses • Facility to store patient information to be used when necessary
	**Public service - e-governance**	• Bill payment • Specimen forms, loan information • Government Information Centre • Ability to apply to State universities through the Internet • To obtain circulars, legal services • Information in Gazette • The map of Sri Lanka
	**e-commerce**	• Exchange of goods through the Internet • Online shopping and purchasing

6.3 The digital divide

Most countries in the world are developing fast. We speak of two kinds of countries: **developed** countries and **developing** countries. ICT helps a society to develop, but ICT has not yet reached some countries. Some countries lag behind.

(i) THE DIGITAL DIVIDE

Some societies in the world have ready access to computers and the Internet, while some others do not. This gap between those who have and those who have not is known as the **digital divide**.



(>) DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 6.1)

ICT services around us. Work in groups of four.

1	Each member of the group takes one of the four fields of 6.2.
2	Ask your family which services of that field they have used.
3	Write the services in the table below, in the column of their field.
4	Tell the class: which field does your family use most?

e-learning	e-health	e-governance	e-commerce

(>) DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 6.2)

The digital divide. Discuss in groups, then write in your exercise book.

1	Choose two services of 6.2, for example paying a bill.
2	Write how a family **with** ready access to ICT gets each service.
3	Write how a family **without** it gets the same service.
4	Discuss: who saves more time? Why do some countries lag behind?

A Which field?

8 marks

Which field of ICT services is it? Tick one box for each line.

It is ...	e-learning	e-health	e-governance	e-commerce
A teacher explains the lesson with pictures and video clips.	☐	☐	☐	☐
Nimal applies to a State university through the Internet.	☐	☐	☐	☐
A doctor finds an illness with computer based modern equipment.	☐	☐	☐	☐
Nimali buys a school bag from an online shop.	☐	☐	☐	☐
Father finds information in the Gazette on the Internet.	☐	☐	☐	☐
Our school sends messages to another school.	☐	☐	☐	☐
A farmer exchanges goods through the Internet.	☐	☐	☐	☐
A hospital stores patient information to use when necessary.	☐	☐	☐	☐

B Fill in the blanks

8 marks

Use the words in the box. Two words are extra.

WORD BANK

e-learning	efficiently	anywhere	patient	e-governance
shopping	digital divide	behind	slowly	Gazette

1. Teaching using ICT and the Internet is called ..... .
2. ICT has helped us to complete tasks more ..... and quickly.
3. With e-learning, we can learn anytime from ..... .
4. A hospital uses ICT to store ..... information.
5. Bill payment and specimen forms are services of ..... .
6. e-commerce includes online ..... and purchasing.
7. The gap between those who have ready access to ICT and those who have not is the ..... .
8. ICT has not yet reached some countries. Some countries lag ..... .

"Technology is at its best when it brings people together."

- THE KING OF ICT

C Write two services

8 marks

Write two services of each field. One mark for each service.

Field	Two services we get with ICT
School e-learning	1. ..... 2. .....
Hospital e-health	1. ..... 2. .....
Public service - e-governance	1. ..... 2. .....
e-commerce	1. ..... 2. .....

D Choose the correct answer

5 marks

Underline your answer.

1. Teaching using ICT and the Internet is
   (a) e-commerce (b) e-learning (c) e-health (d) e-governance

2. Which is a service of Hospital e-health?
   (a) bill payment (b) online shopping
   (c) storing patient information (d) the map of Sri Lanka

3. Exchange of goods through the Internet is part of
   (a) e-commerce (b) e-learning
   (c) e-health (d) the digital divide

4. Which of these is NOT a service of e-governance?
   (a) information in Gazette (b) specimen forms
   (c) circulars and legal services (d) online shopping

5. Some countries lag behind because
   (a) ICT has not yet reached them (b) they have too many computers
   (c) they use e-commerce (d) they have no roads

E True or false

6 marks

Write T (true) or F (false) in the box.

1.		ICT has brought about great changes everywhere, in every field.
2.		e-learning means learning only from printed books.
3.		Bill payment is a service of e-governance.
4.		Hospitals cannot use computers to diagnose illnesses.
5.		Every society in the world has ready access to computers and the Internet.
6.		Online shopping and purchasing are part of e-commerce.

CHAPTER 2

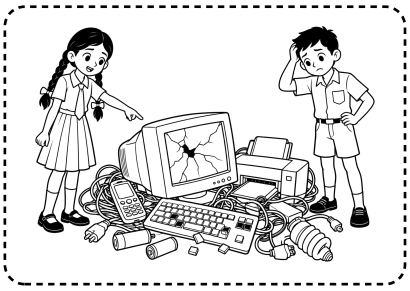
E-waste and the 3R Technique

GARBAGE EVERYWHERE!

"Look! There is garbage everywhere. The whole environment is dirty!"

"Electronic waste is the most dangerous. It harms people too."

Meet ten harmful e-waste items, their toxic poisons, and the 3R technique.



Colour me!

6.4 Harmful electronic waste

Old and broken electronic items that are thrown away are **electronic waste** (**e-waste**). Of all the garbage, e-waste is the most dangerous: it makes the environment dirty, and it also harms people. These are a few harmful e-waste items:

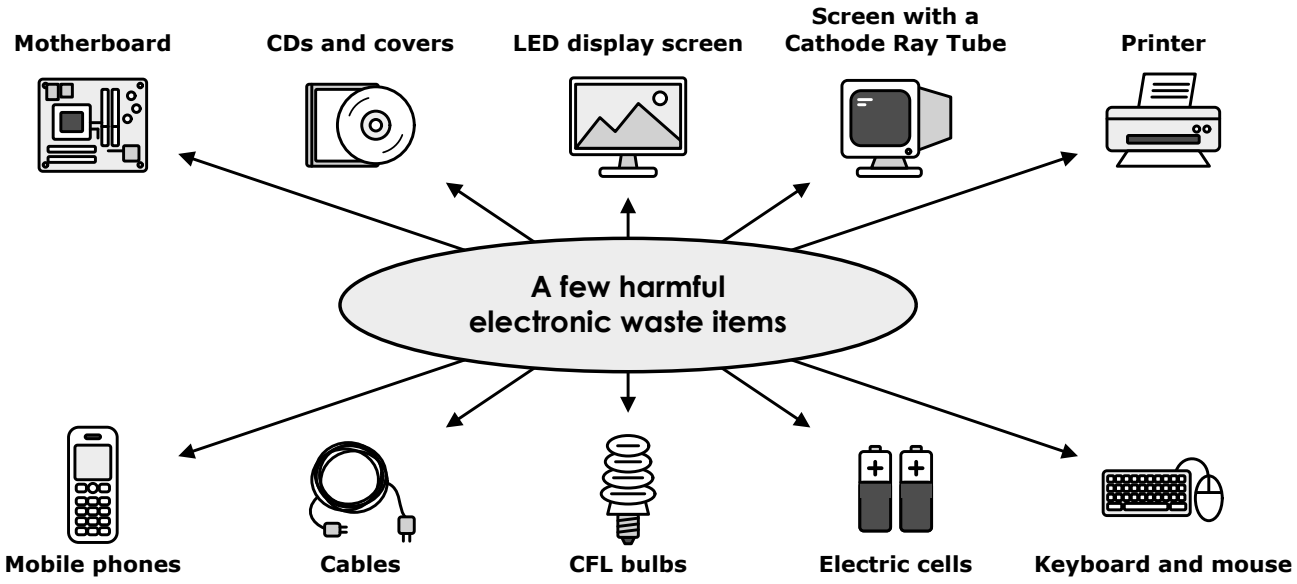


Figure 6.1 - e-waste (textbook page 78)

DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 6.3)

Find the e-waste. Do this at home and at school.

1	Look for old or broken electronic items at home and at school.
2	Write their names. Which of them are in Figure 6.1?
3	Find out what happens to them now: are they kept, thrown away or burnt?
4	Never open, break or burn them. Never touch a broken CFL bulb.

6.5 E-waste harms our health

E-waste releases **toxic poisons** that are likely to harm humans (textbook Figure 6.2):

Toxic substance	How it can harm people
Selenium	loss of hair, nails; can cause allergies
Beryllium	lung cancer, breathing difficulties
Mercury	affects nerves, liver
Chromium	various allergies
Arsenic	cancer, nerve problems, skin diseases
Trichloroethylene	affects liver, gall bladder; infant deaths
Cadmium	kidney weaknesses; affects bone density; causes cancer
Lead	defects in brain, stomach and blood circulation
Polyvinyl Chloride	strong gas when these plastics are burnt: breathing related diseases
Barium	swelling of brain, weak muscles, damage to heart
Brominated Flame Retardants	physical weakness
Polychlorinated biphenyls	destroys liver, respiratory system; causes cancer
Dioxins and Furans	skin diseases, weak nerves, cancer

(!) NEVER BURN OR DUMP E-WASTE

Improper disposal of e-waste is a threat to all living beings and to the environment.

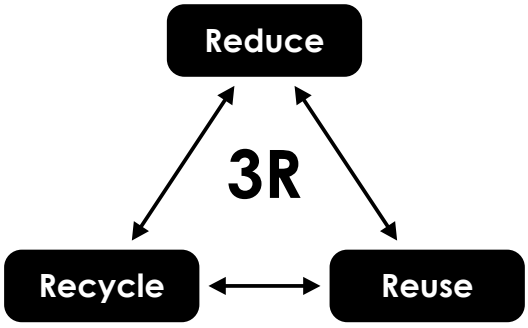
(>) DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 6.4)

A poster: e-waste harms our health. Work in groups.

1	Choose four toxic substances from the table of 6.5.
2	Draw a big outline of a person on a sheet of paper.
3	Draw a line from each substance to the part of the body it harms.
4	Write the harm, and why we must never burn e-waste. Show the poster.

6.6 The 3R technique

We can minimize e-waste through the **3R technique**: **R**educe, **R**euse and **R**ecycle. The 3R concept can be used for the **proper disposal** of e-waste.



R	What we do (textbook page 80)
Reduce	Minimize the purchase or collection of unnecessary goods.
Reuse	Use it again and again after repairing it, wherever possible.
Recycle	Hand it over to suitable agencies for recycling purposes.

DO IT - PRACTICAL 5 (WORKBOOK ACTIVITY 6.5)

A 3R plan for our school. Do this with your teacher.

1	List the old electronic items in your school computer lab or office.
2	For each one decide: **Reuse** it (repair it) or **Recycle** it?
3	Write one rule for **Reduce**: do not buy goods that are not needed.
4	With your teacher, find a suitable agency that recycles e-waste.

A Name the e-waste

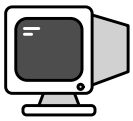
8 marks

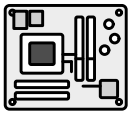
Write the name of each item. Use the names in the box. Two names are extra.

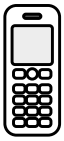
E-WASTE ITEMS

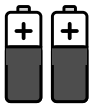
Cables	CFL bulbs	Electric cells	Motherboard	Keyboard and mouse
Printer	Mobile phones	LED display screen	CDs and covers	Screen with a Cathode Ray Tube

1 

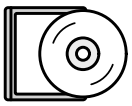
2 

3 

4 

5 

6 

7 

8 

B Match the poison

6 marks

Write the correct letter in the box.

A - Toxic substance		B - How it harms people
1. Selenium		**A.** various allergies
2. Mercury		**B.** destroys the liver and the respiratory system; causes cancer
3. Arsenic		**C.** loss of hair and nails
4. Chromium		**D.** physical weakness
5. Brominated Flame Retardants		**E.** cancer, nerve problems, skin diseases
6. Polychlorinated biphenyls		**F.** affects the nerves and the liver

C Which part of the body?

6 marks

Which part of the body does it harm? Tick one box (Figure 6.2).

It harms ...	Brain	Lungs	Liver	Kidneys
Lead	☐	☐	☐	☐
Beryllium	☐	☐	☐	☐
Trichloroethylene	☐	☐	☐	☐
Cadmium	☐	☐	☐	☐
Barium	☐	☐	☐	☐
Polyvinyl Chloride (burnt plastics)	☐	☐	☐	☐

D Reduce, reuse or recycle?

8 marks

Which R of the 3R technique is it? Tick one box.

What we do	Reduce	Reuse	Recycle
Do not buy a new phone while the old one works.	☐	☐	☐
Repair the broken keyboard and use it again.	☐	☐	☐
Hand over old mobile phones to a recycling agency.	☐	☐	☐
Buy only the cables that you really need.	☐	☐	☐
Repair the old printer and use it again.	☐	☐	☐
Give dead electric cells to a recycling agency.	☐	☐	☐
Do not collect goods that you do not need.	☐	☐	☐
Use the same CD again and again.	☐	☐	☐

E True or false

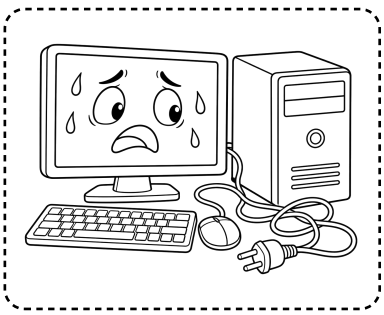
6 marks

Write T (true) or F (false) in the box.

1. ☐ E-waste makes the environment dirty, but it does not harm people.
2. ☐ Mobile phones and cables are harmful e-waste items.
3. ☐ Beryllium can cause lung cancer and breathing difficulties.
4. ☐ Burning plastics is a safe way to get rid of e-waste.
5. ☐ Reuse means handing e-waste over to suitable agencies.
6. ☐ The 3R concept can be used for the proper disposal of e-waste.

"A clean world is a healthy world. Never burn your e-waste."

- THE KING OF ICT



Colour me!

F Fill in the blanks

8 marks

Use the words in the box. Two words are extra.

WORD BANK

dangerous	environment	Tube	CFL	Lead
Reduce	Reuse	agencies	Recycle	Mercury

1. Of all the garbage, electronic waste is the most ..... .
2. E-waste makes the ..... dirty, and it also harms people.
3. An old screen with a Cathode Ray ..... is a harmful e-waste item.
4. ..... bulbs and electric cells are harmful e-waste items.
5. ..... causes defects in the brain, stomach and blood circulation.
6. Minimize the purchase of unnecessary goods. This is ..... .
7. Use it again and again after repairing it. This is ..... .
8. Hand over e-waste to suitable ..... for recycling purposes.

G Choose the correct answer

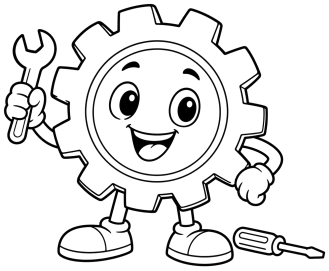
5 marks

Underline your answer.

1. Of all the garbage, the most dangerous waste is
   (a) paper waste (b) food waste (c) electronic waste (d) garden waste
2. Which of these is a harmful e-waste item?
   (a) a wooden desk (b) a CFL bulb (c) a glass bottle (d) a cotton shirt
3. Dioxins and Furans can cause
   (a) skin diseases, weak nerves, cancer (b) loss of hair and nails
   (c) kidney weaknesses (d) swelling of the brain
4. The 3R technique is
   (a) Read, Write, Repeat (b) Reduce, Reuse, Recycle
   (c) Repair, Return, Reset (d) Run, Rest, Relax
5. Improper disposal of e-waste is a threat to
   (a) only computers (b) only plants
   (c) all living beings and the environment (d) nobody

"Reduce, reuse, recycle: three small words that protect our planet."

- THE KING OF ICT



Colour me!

H Word search

10 marks

Find the 10 toxic substances. They go across, down and slanting.

R	P	U	T	T	N	C	F	S	T	R	U
I	T	F	U	R	A	N	S	N	C	Y	A
C	U	B	E	R	Y	L	L	I	U	M	M
H	U	L	E	A	D	O	T	X	K	S	E
R	S	T	S	Y	C	Y	U	O	T	K	R
O	A	C	H	E	E	D	D	I	R	P	C
M	R	B	R	Y	L	B	W	D	A	M	U
I	S	H	P	H	M	E	U	W	R	H	R
U	E	B	W	W	C	W	N	B	O	O	Y
M	N	M	U	I	R	A	B	I	U	C	C
G	I	B	W	F	C	C	A	M	U	W	R
T	C	A	D	M	I	U	M	T	R	M	A

FIND THESE WORDS

- ☐ LEAD
- ☐ MERCURY
- ☐ CADMIUM
- ☐ ARSENIC
- ☐ BARIUM
- ☐ SELENIUM
- ☐ CHROMIUM
- ☐ BERYLLIUM
- ☐ DIOXINS
- ☐ FURANS

CHAPTER 3

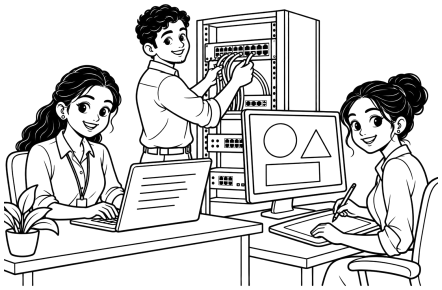
Computer Related Jobs

? WILL ICT GET ME A JOB?

"Will learning about ICT help us to get a job easily?"

"Yes! A proper education on ICT related subjects gives ample job opportunities."

Meet ten computer related jobs and what each person does.



6.7 Computer related job opportunities

Those who have obtained a proper education on ICT related subjects have **ample job opportunities**: there are many jobs to choose from. These are some computer related jobs and their responsibilities (textbook page 81):

	Job	Responsibilities - what the person does
1	**Software Quality Assurance Engineer**	Carries out tasks to assure that the software functions properly.
2	**Software Engineer**	Designs and develops software.
3	**Software Architect**	Designs the architecture of software (its components and their interconnections).
4	**Programmer**	Develops computer programs using programming languages.
5	**Network Administrator**	Installs, configures and maintains computer networks.

	Job	Responsibilities - what the person does
6	**Database Administrator**	Plans, installs, configures and maintains databases.
7	**Web Application Developer**	Designs, develops and maintains websites.
8	**Graphic Designer**	Develops web pages, advertisements, magazines and banners using software.
9	**System Analyst**	Analyses organizational needs and designs computer systems; coordinates the development of software.
10	**IT Consultant**	Advises organizations on the matters related to information technology.

DO IT - PRACTICAL 6 (WORKBOOK ACTIVITY 6.6)

ICT jobs in the newspaper. Work in pairs.

1	Collect the job advertisements of a few newspapers.
2	Find the advertisements for computer related jobs.
3	Write each job title and its responsibilities in the table below.
4	Is the job in the table of 6.7? Which ICT job is asked for most?

Job title	Responsibilities	In the table of 6.7?

DO IT - PRACTICAL 7 (WORKBOOK ACTIVITY 6.7)

Meet an ICT worker. Do this with your family or your teacher.

1	Choose the computer related job that you like most (6.7).
2	Talk to a person who works with computers. Ask what he or she does.
3	Ask which ICT related subjects he or she studied for the job.
4	Write a short report, **My dream ICT job**, and read it to the class.

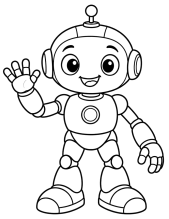
The job I like: _____

What the person does: _____

What he or she studied: _____

"Learn ICT well today, and many doors will open for you tomorrow."

- THE KING OF ICT



Colour me!

A Match the jobs

8 marks

Write the correct letter in the box.

A - Job		B - What the person does
1. Software Quality Assurance Engineer		**A.** Plans, installs, configures and maintains databases.
2. Software Architect		**B.** Advises organizations on matters related to information technology.
3. Network Administrator		**C.** Designs the architecture of software: its components and their interconnections.
4. Database Administrator		**D.** Develops web pages, advertisements, magazines and banners using software.
5. Web Application Developer		**E.** Carries out tasks to assure that the software functions properly.
6. Graphic Designer		**F.** Installs, configures and maintains computer networks.
7. System Analyst		**G.** Analyses organizational needs and designs computer systems.
8. IT Consultant		**H.** Designs, develops and maintains websites.

B Who should do it?

6 marks

Write the job of the person who should do it (6.7).

The problem	Who should do it?
The computer network of the school does not work.	
A shop needs banners and advertisements.	
A bank needs a person to look after its databases.	
New software must be checked to work properly.	
A team needs a person to write computer programs.	
A hospital wants advice on information technology.	

C True or false

6 marks

Write T (true) or F (false) in the box.

1. ☐ People with a proper education on ICT related subjects have ample job opportunities.

2. ☐ A Graphic Designer installs and maintains computer networks.

3. ☐ A Programmer develops computer programs using programming languages.

4. ☐ A Database Administrator designs banners and magazines.

5. ☐ An IT Consultant advises organizations on matters related to information technology.

6. ☐ A Software Architect only repairs keyboards and mice.

D Choose the correct answer

5 marks

Underline your answer.

1. People with a proper education on ICT related subjects have
   (a) no jobs (b) ample job opportunities
   (c) only one job (d) jobs only in shops

2. Who designs and develops software?
   (a) a Software Engineer (b) a Graphic Designer
   (c) an IT Consultant (d) a Network Administrator

3. Who analyses organizational needs and designs computer systems?
   (a) a Programmer (b) a System Analyst
   (c) a Graphic Designer (d) a Database Administrator

4. A Web Application Developer designs, develops and maintains
   (a) printers (b) websites (c) cables (d) banks

5. The components of software and their interconnections are designed by
   (a) a Software Architect (b) a Programmer
   (c) a Graphic Designer (d) an IT Consultant

E My dream ICT job

4 marks

Think about the jobs of 6.7. Write your answers.

1. Which computer related job would you like to do? (1 mark)

2. What does a person in that job do? (2 marks)

3. Why must you study ICT related subjects well for this job? (1 mark)

F Answer these

4 marks

Write full sentences.

1. What does a System Analyst do? Write two things. (2 marks)

2. What is the difference between a Programmer and a Software Architect? (2 marks)

G Read the advertisement

5 marks

You find this advertisement in a newspaper (Practical 6).

> **WANTED - A NETWORK ADMINISTRATOR.** Green Valley Bank needs a Network Administrator for its head office in Kandy. You will install, configure and maintain the computer network of the bank. You must have a proper education on ICT related subjects. Apply before 30th November.

1. Which job does the advertisement offer? (1 mark)

.....

2. What will this person do at the bank? (2 marks)

.....

.....

3. What education must this person have? (1 mark)

.....

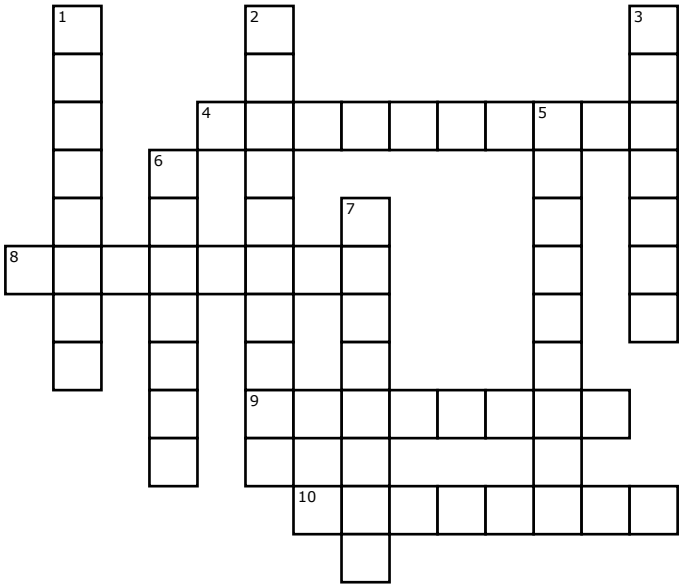
4. The bank also needs a person to plan, install and maintain its databases. Name that job. (1 mark)

.....

H Crossword

10 marks

All the words are about the jobs of 6.7. One letter in each square.



ACROSS

4. IT ... : advises organizations on IT matters (10)
8. ... Administrator: plans, installs and maintains databases (8)
9. Software ... : designs and develops software (8)
10. A Web Application Developer designs, develops and maintains ... (8)

DOWN

1. A Quality Assurance Engineer checks that it functions properly (8)
2. Develops computer programs using programming languages (10)
3. ... Administrator: installs and maintains computer networks (7)
5. Software ... : designs the architecture of software (9)
6. System ... : analyses organizational needs (7)
7. Graphic ... : makes advertisements, magazines and banners (8)

REVISION

Unit 6 in One Page

Read this the night before the test

Topic	The key facts
ICT and society	Great changes in every field: ICT helps us to complete tasks more efficiently and quickly.
School e-learning	Teaching using ICT and the Internet: pictures and video clips, information from the Internet, messages with other schools, learning anytime from anywhere.
Hospital e-health	Computer based equipment to diagnose illnesses; patient information stored for later.
e-governance	Bill payment, specimen forms, loan information, Government Information Centre, applying to State universities, circulars, legal services, Gazette, map of Sri Lanka.
e-commerce	Exchange of goods through the Internet; online shopping and purchasing.
Digital divide	The gap between societies with ready access to computers and the Internet and those without it.
e-waste	The most dangerous waste: it makes the environment dirty and it harms people.
Harmful items	Motherboard, CDs and covers, LED and CRT screens, printer, keyboard and mouse, electric cells, CFL bulbs, cables, mobile phones.
Toxic poisons	Lead: brain. Mercury: nerves, liver. Cadmium: kidneys. Beryllium: lungs. Barium: brain, heart. Arsenic, Dioxins and Furans: skin diseases, cancer.
3R technique	Reduce: fewer unnecessary goods. Reuse: repair it and use it again. Recycle: hand it over to suitable agencies.
ICT jobs	Ample jobs for people with a proper education on ICT related subjects: Software Engineer and Architect, Programmer, Quality Assurance Engineer, System Analyst, Database and Network Administrators, Web Application Developer, Graphic Designer, IT Consultant.

★ TICK WHEN YOU CAN DO IT

- ☐ I can explain how ICT has changed our society and what e-learning is.
- ☐ I can name ICT services of schools, hospitals, the public service and shops.
- ☐ I can explain the digital divide.
- ☐ I can name ten harmful e-waste items.
- ☐ I can say how the toxic poisons in e-waste harm people.
- ☐ I can explain Reduce, Reuse and Recycle.
- ☐ I can name computer related jobs and say what each person does.

WORD POWER

Unit 6 Word List

Twenty-six words you must know

Word	What it means
ICT	Information and Communication Technology.
Society	The people who live together in a community or a country.
Efficiency	Doing a task well and quickly, without wasting time or work.
e-learning	Teaching (and learning) using ICT and the Internet.
e-health	ICT in hospitals: finding illnesses with computer based equipment, storing patient information.
e-governance	Public services through ICT: bill payment, forms, the Gazette, applying to universities ...
e-commerce	Exchange of goods through the Internet; online shopping and purchasing.
Gazette	The official government publication with public information.
Digital divide	The gap between those who have ready access to computers and the Internet and those who have not.
Developing countries	Countries that are still developing. ICT has not yet reached some of them.
e-waste	Electronic waste: old or broken electronic items that are thrown away.
Motherboard	The main circuit board of a computer.
Cathode Ray Tube	The big glass tube of an old, box-shaped screen.
CFL bulb	Compact fluorescent lamp: a bulb with a spiral tube.
Electric cell	A small battery that gives electricity.
Toxic	Poisonous: harmful to people and to all living beings.
Lead	A toxic substance in e-waste: defects in the brain, stomach and blood circulation.
Mercury	A toxic substance in e-waste: it affects the nerves and the liver.
Cadmium	A toxic substance in e-waste: kidney weaknesses; it affects bone density; cancer.
3R technique	Reduce, Reuse, Recycle: the way to minimize e-waste.
Reduce	Minimize the purchase or collection of unnecessary goods.
Reuse	Use again and again after repairing, wherever possible.
Recycle	Hand over to suitable agencies for recycling purposes.
Programmer	Develops computer programs using programming languages.
System Analyst	Analyses organizational needs and designs computer systems.
IT Consultant	Advises organizations on matters related to information technology.

"Learn the words today, and you will speak like an expert tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 6 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. The use of ICT has helped us to complete tasks more
   (a) slowly (b) efficiently (c) badly (d) rarely
2. Hospitals use ICT to store patient
   (a) information (b) food (c) beds (d) money
3. We speak of two kinds of countries: developed countries and
   (a) developing (b) big (c) old (d) cold
4. Which of these is NOT a harmful e-waste item?
   (a) a printer (b) a cable (c) a mouse (d) a ruler
5. The 3R technique is used to minimize
   (a) e-waste (b) e-learning (c) e-health (d) e-commerce
6. A Software Engineer designs and develops
   (a) software (b) banners (c) bridges (d) houses
7. In the field of ICT, qualified people have many
   (a) jobs (b) problems (c) holidays (d) games
8. A - Some countries lag behind in ICT. B - Cables are not e-waste. C - A Software Architect designs the architecture of software. Which are correct?
   (a) A only (b) A and C (c) B and C (d) A, B and C

2 Fill in the blanks

8 marks

One word or a few words in each blank.

1. ICT has not yet reached some countries. Some countries lag ..... .
2. A hospital uses computer based modern ..... to diagnose illnesses.
3. Of all the garbage, ... waste is the most dangerous.
4. Electric cells, cables and ..... phones are harmful e-waste items.
5. A ..... develops computer programs using programming languages.
6. A ..... Administrator plans, installs, configures and maintains databases.
7. A System ..... analyses organizational needs and designs computer systems.
8. A ..... Application Developer designs, develops and maintains websites.

3 Name the pictures

8 marks

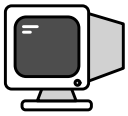
Cards 1 - 4: name the field of ICT services. Cards 5 - 8: name the e-waste item.

1 

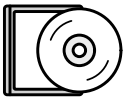
2 

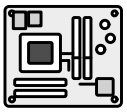
3 

4 

5 

6 

7 

8 

4 Answer these

16 marks

Write in the space given.

1. What is e-learning? Write one way it helps students. (2 marks)

2. Write two services of Public service - e-governance. (2 marks)

3. What is the digital divide? (2 marks)

4. Why is electronic waste the most dangerous waste? (2 marks)

5. Name two toxic substances in e-waste and the harm that each one does. (2 marks)

6. Explain the 3R technique. (3 marks)

7. Write what each of these people does: a Network Administrator, a Graphic Designer and an IT Consultant. (3 marks)